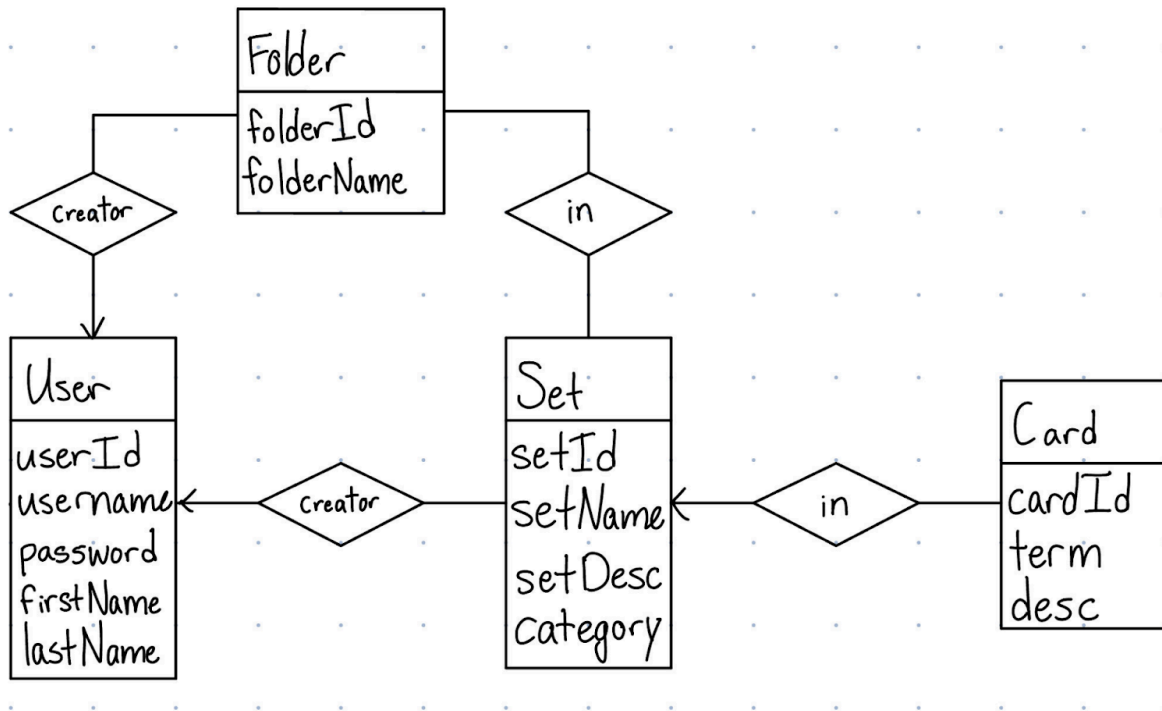


Group 3 - Term Project Database Design

Preliminary ER diagram:



Database Entities: Users, Sets, Flashcards, Folders, Set_Folders

Users(userId, username, password, firstName, lastName)

Sets(setId, name, description, category)

Cards(cardId, cardTerm, cardDesc)

Folders(folderId, folderName)

Set_Folders(setId, folderId)

Functional Dependencies:

Users: {
 userId -> username, password, firstName, lastName
}

Sets: {
 cardId -> setId, cardTerm, cardDesc
}

Cards: {
 cardId -> setId, cardTerm, cardDesc
}

Folders: {
 folderId -> userId, folderName
}

Set_Folders: {
 setId, folderId -> setId, folderId
}

The relations are already in 3NF.

Final tables:

Users (
 userId int auto_increment,
 username varchar(255) not null,
 password varchar(255) not null,
 firstName varchar(255) not null,
 lastName varchar(255) not null,
 primary key (userId),
 unique (username),
)

Sets (

```
    setId int auto_increment,  
    userId int not null,  
    setName varchar(255) not null,  
    setDescription varchar(500) not null,  
    setCategory varchar(255) not null,  
    primary key(setId)  
    foreign key(userId) references Users (userId) on delete cascade  
    foreign key(folderId) references Folders (folderId) on delete cascade
```

)

Flashcards (

```
    cardId int not null,  
    setId int not null,  
    cardTerm varchar(255) not null,  
    cardDesc varchar(500) not null,  
    primary key(cardNum)  
    foreign key(setId) references Sets (setId)
```

)

Folders (

```
    folderId int not null,  
    userId int not null,  
    folderName varchar(255),  
    primary key(folderId),  
    foreign key(userId) references Users (userId)
```

)

Set_Folders (

```
    setId int not null,  
    folderId int not null,  
    primary key(setId, folderId),  
    foreign key(setId) references Sets(setId),  
    foreign key(folderId) references Folders(folderId)  
)
```